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Exercise. Duality and Sensitivity Analysis

**Problem 1.** Consider the following linear program:

$$\begin{array}{lll} \text{maximize} & 3x_1 + 4x_2 + 3x_3 + 6x_4 & \\ \text{subject to} & 2x_1 + x_2 - x_3 + x_4 & \geq 12 \\ & x_1 + x_2 + x_3 + x_4 & = 8 \\ & -x_2 + 2x_3 + x_4 & \leq 10 \\ & x_1, x_2, x_3, x_4 & \geq 0. \end{array} \tag{1}$$

After transforming the problem into standard form and apply Simplex method, we obtain the final tableau as follows:

| B | 0 | 2 | 9 | 0 | 3 | 0 | 36 |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | −2 | 0 | −1 | 0 | 4 |
| 4 | 0 | 1 | 3 | 1 | 1 | 0 | 4 |
| 6 | 0 | −2 | −1 | 0 | −1 | 1 | 6 |

a) Derive the dual problem of the linear program (1) and calculate a dual solution based on complementarity conditions. Given that the optimal solution to the primal solution is unique, investigate whether the dual solution is unique.

b) Do the optimal solution and the objective function value change if we

- decrease the objective function coefficient for $x_3$ to 0?
- increase the objective function coefficient for $x_3$ to 9?
- decrease the objective function coefficient for $x_4$ to 5?
- increase the objective function coefficient for $x_1$ to 7?

c) Find the possible range for adjusting the coefficient 8 of the second constraint such that the current basis is kept optimal.

**Solution 1.**

a) The dual of problem (1) is given by

$$\begin{array}{lll} \text{minimize} & 12y_1 + 8y_2 + 10y_3 & \\ \text{subject to} & y_1 \leq 0, y_2 \text{ free}, y_3 \geq 0 & \\ & 2y_1 + y_2 & \geq 3 \\ & y_1 + y_2 - y_3 & \geq 4 \\ & -y_1 + y_2 + 2y_3 & \geq 3 \\ & y_1 + y_2 + y_3 & \geq 6. \end{array}$$

Using the complementarity conditions, we can infer $2y_1 + y_2 - 3 = 0$, $y_1 + y_2 + y_3 - 6 = 0$, and $y_3(-x_2^* + 2x_3^* + x_4^* - 10) = -6y_3 = 0$. (Since the optimal slack variables are not all zero, one of the primal constraints needs to be inactive). This yields $y_3 = 0$ and:

$$2y_1 + y_2 = 3, \quad y_1 + y_2 = 6 \quad \Longrightarrow \quad y_1 = -3, \quad y_2 = 9.$$

Since the primal solution is unique and the complementarity conditions fully characterize the dual solution $y^* = (-3, 9, 0)^\top$, the dual problem has a unique solution as well.

b) Using the final simplex tableau, we obtain:

$$A_B^{-1}A_N = \begin{pmatrix} 0 & -2 & -1 \\ 1 & 3 & 1 \\ -2 & -1 & -1 \end{pmatrix} \quad \text{and we have} \quad A_B^{-1} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ 1 & -2 & 1 \end{pmatrix}.$$

We now discuss the different questions step by step:

- Decreasing $c_3 = 3$ to 0 means that the new costs (in standard form) are given by: $\tilde{c} = (-3, -4, 0, -6, 0, 0)^\top$. Due to $\{3\} \notin B$, we can simply check:

$$r_N^\top + \begin{pmatrix} 0 & 3 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 12 & 3 \end{pmatrix} \geq 0.$$

  In this case, both the optimal value as well as the optimal solution do not change.

- Increasing $c_3 = 3$ to 9 means that the new costs (in standard form) are given by: $\tilde{c} = (-3, -4, -9, -6, 0, 0)^\top$. Due to $\{3\} \notin B$, we can simply check:

$$r_N^\top + \begin{pmatrix} 0 & -6 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 3 \end{pmatrix} \geq 0.$$

  In this case, both the optimal value as well as the optimal solution do not change.

- Decreasing $c_4 = 6$ to 5 means that the new costs (in standard form) are given by: $\tilde{c} = (-3, -4, -3, -5, 0, 0)^\top$. We then have:

$$\tilde{c}_N^\top - \tilde{c}_B^\top A_B^{-1} A_N = \begin{pmatrix} -4 & -3 & 0 \end{pmatrix} - \begin{pmatrix} -3 & -5 & 0 \end{pmatrix} \begin{pmatrix} 0 & -2 & -1 \\ 1 & 3 & 1 \\ -2 & -1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 6 & 2 \end{pmatrix}.$$

  Thus, $x^* = (4, 0, 0, 4, 0, 6)^\top$ will remain optimal solution with new optimal value 32.

- Increasing $c_1 = 3$ to 7 means that the new costs (in standard form) are given by: $\tilde{c} = (-7, -4, -3, -6, 0, 0)^\top$. We then have:

$$\tilde{c}_N^\top - \tilde{c}_B^\top A_B^{-1} A_N = \begin{pmatrix} -4 & -3 & 0 \end{pmatrix} - \begin{pmatrix} -7 & -6 & 0 \end{pmatrix} \begin{pmatrix} 0 & -2 & -1 \\ 1 & 3 & 1 \\ -2 & -1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & -1 \end{pmatrix}.$$

  The optimal solution will change in this case. (Calling CVX, we can find $x^* = (8, 0, 0, 0, 4, 10)^\top$ with optimal value 56).

c) We need to find the range of $\lambda$ such that $x_B^* + \lambda A_B^{-1} e_2$ is nonnegative, i.e.,

$$\begin{pmatrix} 4 \\ 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix} = \begin{pmatrix} 4 - \lambda \\ 4 + 2\lambda \\ 6 - 2\lambda \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

This yields $-2 \leq \lambda \leq 3$ and thus, the coefficient 8 can be chosen from the interval $[6, 11]$ without changing the current optimal basis.

**Problem 2 (50pts).** An insurance company is introducing three products: special risk insurance, mortgage insurance, and long-term care insurance. The expected profit is \$500 per unit on special risk insurance, \$250 per unit on mortgage insurance and \$600 per unit on long term care insurance. The work requirements are as follows:

The management team wants to establish sales quotas for each product to maximize the total expected profit.

| Department | Working hours per unit | | | Working hours available |
|---|---|---|---|---|
| | Special risk | Mortgage | Long-term care | |
| Underwriting | 2 | 1 | 1 | 240 |
| Administration | 3 | 1 | 2 | 150 |
| Claims | 1 | 2 | 4 | 180 |

a) Formulate this problem as a linear optimization problem. Specify the decision variables, objective function, and constraints.

b) After solving the problem, the final simplex tableau (for the standard form) is given as below (the variables are in the natural order as in the description of the problem):

| B | 0 | 50 | 0 | 0 | 140 | 80 | 35400 |
|---|---|---|---|---|---|---|---|
| 4 | 0 | 0.5 | 0 | 1 | -0.7 | 0.1 | 153 |
| 1 | 1 | 0 | 0 | 0 | 0.4 | -0.2 | 24 |
| 3 | 0 | 0.5 | 1 | 0 | -0.1 | 0.3 | 39 |

Show the dual variables corresponding to the services of the three departments. Using complementarity conditions to explain why mortgage insurance is not sold.

c) Find the range of working hours available for underwriting to keep the current basis optimal.

d) Find the range of the expected profit on long-term care insurance such that the current basis remains optimal.

**Solution 2.**

a) Let $x_i$ idenote the number of $i$th insurance product. The linear program is

$$\begin{aligned} \text{maximize} \quad & 500x_1 + 250x_2 + 600x_3 \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 \leq 240 \\ & 3x_1 + x_2 + 2x_3 \leq 150 \\ & x_1 + 2x_2 + 4x_3 \leq 180 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

b) From the final tableau, we see that the shadow prices are \$0, \$140, \$80 respectively. If the company sells one unit of mortgage, it requires 1 hour for underwriting, 1 hour for administration, and 2 hours for claims. So, the cost to sell 1 unit of mortgage is $140 + 2 \times 80 = 300$. However, the profit for one unit of mortgage is only \$250 per unit. Therefore, the company cannot make any net profit by selling mortgage.

c) We have $B = \{4, 1, 3\}$ ,and we need to determine the range of $\lambda$ such that

$$\bar{b} + \lambda A_B^{-1} e_1 \geq 0$$

$A_B^{-1}$ is the last three columns of the final tableau, so

$$A_B^{-1} = \begin{pmatrix} 1 & -0.7 & 0.1 \\ 0 & 0.4 & -0.2 \\ 0 & -0.1 & 0.3 \end{pmatrix}$$

It is then straightforward to verify that we must have $-153 \leq \lambda$, and therefore $b_1 \geq 87$.

d) The optimal basis remains the same if and only if new reduced cost is nonnegative, i.e., $\bar{r}_N - \lambda A_N^T \left(A_B^{-1}\right)^T e_3 \geq 0$

$$\begin{cases} 50 - 0.5\lambda \geq 0 \\ 140 + 0.1\lambda \geq 0 \\ 80 + 0.3\lambda \geq 0 \end{cases}$$

We can find $A_B^{-1} A_N$ from the final tableau, and it is therefore straightforward to verify that $-1400 \leq \lambda \leq 100$, and therefore that $c_3 \in [500, 2000]$. (However, the problem might have multiple optimal solutions when $c_3 = 500$ or 2000 - as demonstrated by running CVX).